



COMPONENTS

Air quality sensor

The ZMOD4510 Outdoor Air Quality (OAQ) gas sensor platform now comes with an IP67-qualified waterproof package and a new AI based algorithm that enables ultra-low power selective ozone measurements while maintaining



power consumption under 200µW. The enhanced ZMOD4510 is a miniature digital OAQ solution with selective ozone measurement capabilities, offering visibility into the air quality in users' immediate environments, thus allowing devices such as smart watches, phones, and smoke detectors to monitor harmful ozone gasses typically found outdoors but which can drift indoors.

Renesas Electronics Corporation
www.renesas.com

Power management ICs

The AEM30330 and AEM10330 PMICs utilise a buck-boost architecture, enabling a greater coverage range of input voltage levels (100mV to 4.5V). This means that low and high power sources can be addressed. AEM30330 is targeted at RF/vibration-oriented deployments (including industrial equipment, home automation systems, healthcare monitoring, and household appliances), while AEM10330 is specifically intended for solar deployments (with the scope to manage up to 7 connected photovoltaic cells).

e-peas
www.e-peas.com

Transient voltage suppressor

ESDCAN03-2BM3Y dual automotive-qualified low-capacitance transient voltage suppressor (TVS) protects CAN

and CAN-FD interfaces by delivering all-round performance in a tiny package. The new TVS meets the automotive industry's need for highly miniaturised and high-performing protection devices for the increasing number of high-density electronic control units. By using this device, designers can save 75% or more of the PCB space needed by conventional TVS in SOT23-3L and SOT323-3L packages.

STMicroelectronics
www.st.com

Magnetoresistance speed sensor

The ATS19480 speed sensor IC provides a single-channel solution that's ideal for hybrid and pure electric vehicle transmissions, with use cases extending to two-wheelers, off-road vehicles, and industrial application designs requiring speed-only information.

With an air gap that's 50% larger than existing options, it improves design-in flexibility, expands design margin and tolerance capability, and facilitates a wider range of sensor installation locations, reducing system complexity and boosting efficiency.

Allegro MicroSystems
www.allegromicro.com

Three-phase driver IC

The 6EDL7141 is a fully programmable solution for advanced motor control applications, delivering high power density and improving overall system efficiency. The EiceDRIVER 6EDL7141 features an SPI interface for gate drive out-



puts configuration as well as an integrated power supply and dual charge pumps

to supply all system functions. With an operating voltage of 5.5V to 60V and a configurable driving current up to 1.5A, it can drive a wide range of MOSFETs effectively to fit the application.

Infinion Technologies AG
www.infineon.com

Monolithic microwave IC

The new GMICP2731-10 GaN MMIC power amplifier is designed for use in commercial and defence satellite communications,



5G networks, and other aerospace and defence systems. It is fabricated using GaN-on-Silicon Carbide (SiC) technology and

delivers up to 10W saturated RF output power across the 3.5GHz of bandwidth between 27.5 and 31GHz. Its power-added efficiency is 20%, with 22dB small-signal gain, and 15dB return loss.

Microchip Technology
www.microchip.com

Push-pull transformers

The MID-PPTI series that includes 750317828, 750317829, 750317830, and 750317331 has an isolation voltage of 2,500V AC. The 5V output versions, 750317828 and 750317830, are capable of



handling 250mA current. The 12V output versions, 750317829 and 750317331, can

handle 100mA current. The MID-PPTI transformers are best suited for isolated interface power supply for CAN, RS-485, RS-422, RSS-232, SPI, I2C, low-power LAN, industrial, process control, medical equipment, PLC analogue and digital I/O modules, isolated gate driver power supplies, AC motor drives, and solar inverters.

Würth Elektronik
www.we-online.com

Offline power solution

The new NCP1680 critical conduction mode (CrM) totem-pole PFC controller